Sim

March 16, 2026

[9]:
```
### Needed imports for functions
import numpy as np
from pathlib import Path
from typing import Tuple, Dict, Optional

### Import for plotting
import matplotlib.pyplot as plt

### Importing simulation functions
import PhysComp.sim as sim
```

sim_dataset function simulates a rudimentary azimuthal scattering dataset for given spatial and spectral dimensions. It applied noise based on given constants to mimic real detector data. The function returns the 3-D detector array, 3-D ground truth array and a 1-D param array.

Underlying ground truth calculated by:

0.0.1 Ground Truth $= A_1 sin(x + \phi_1) + A_2 sin(2x + \phi_2) + O$

Where:

A_1 / A_2 = Randomly Generated Amplitudes

ϕ_1 / ϕ_2 **= Randomly Generated Phases**

O = Offset To Ensure Positive Signal

A demonstration of the underlying signal generated using the function is shown in Figure 1a and Figure 2a

Simulated Detector Data calculated by:

0.0.2 Detector Data = Ground Truth + Poission Noise + Gaussian Noise

Where:

Poisson Noise and Gaussian Noise are generated from respective input parameters and numpy functions

A demonstration of the detector data generation is shown below:

[14]:
```
X = 200 # X dimesion size
Y = 200 # Y dimesion size
Bins = 360 # Spectral bins
Possion = 100 # Constant defining possion noise
Gaussian = 100 # Constant defining Gaussian noise
Seed = 123456789 # Generation seed

### Simulating dataset of 200x200 with 360 bins
simmed_data = sim.sim_dataset(X,Y,Bins,Possion,Gaussian, seed = Seed)
```

The rest of the notebook visuals single degree slices in Figure 1 and a single pixel signal in Figure 2

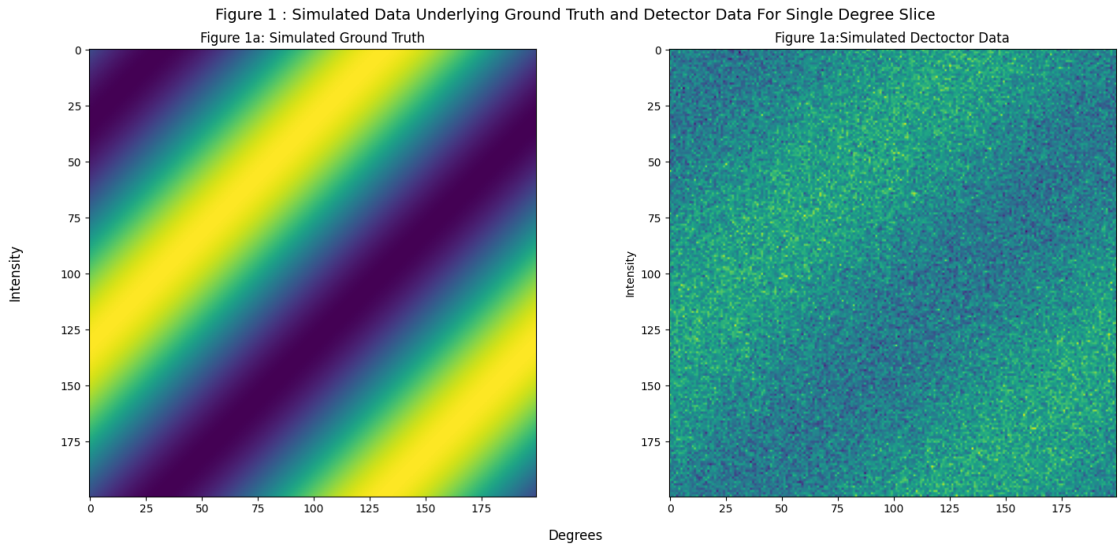
[15]:
```
fig, axes = plt.subplots(1, 2,figsize=(15,7))
plt.suptitle("Figure 1 : Simulated Data Underlying Ground Truth and Detector␣
 ↪Data For Single Degree Slice",fontsize=14)
plt.ylabel("Intensity")

fig.supxlabel("Degrees",fontsize=12)
fig.supylabel("Intensity",fontsize=12)


im0 = axes[0].imshow(simmed_data[1][:,:,0])
axes[0].set_title("Figure 1a: Simulated Ground Truth",fontsize=12)

im1 = axes[1].imshow(simmed_data[0][:,:,0])
axes[1].set_title("Figure 1a:Simulated Dectoctor Data",fontsize=12)

plt.tight_layout()
plt.show()
```



[16]:
```
fig, axes = plt.subplots(1, 2,figsize=(15,7))
plt.suptitle("Figure 2 : Simulated Underlying Ground Truth and Detector For␣
↪Data Single Pixel",fontsize=14)
plt.ylabel("Intensity")

fig.supxlabel("Degrees",fontsize=12)
fig.supylabel("Intensity",fontsize=12)


im0 = axes[0].plot(simmed_data[1][0,0])
axes[0].set_title("Simulated Ground Truth",fontsize=12)
axes[0].grid(True)

im1 = axes[1].plot(simmed_data[0][0][0])
axes[1].set_title("Simulated Detector Data",fontsize=12)
axes[1].grid(True)

plt.tight_layout()
plt.show()
```

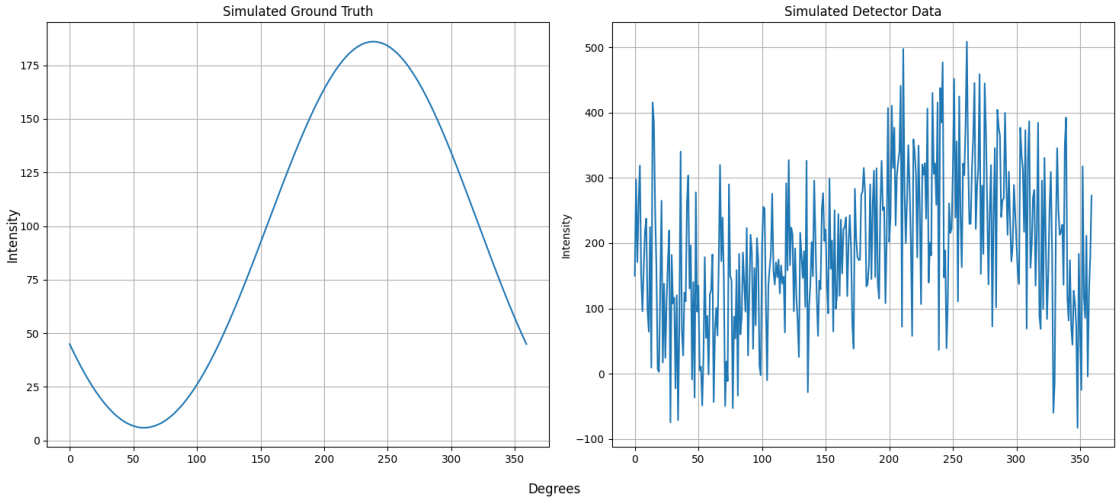



Figure 2 : Simulated Underlying Ground Truth and Detector For Data Single Pixel